# MEDICAL DIAGNOSIS CHATBOT BASED ON LARGE LANGUAGE MODELS

*MINI PROJECT-II REPORT submitted in partial fulfillment of the requirements*

*for the Award of the Degree of*

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**By**


**B.Hanisha - 228W1A1272**

**G.Teja - 228W1A1289**


*Under the Guidance of*

**Dr. M. Suneetha, Ph.D.,**

**Professor and Head**


**DEPARTMENT OF INFORMATION TECHNOLOGY**

**V R SIDDHARTHA ENGINEERING COLLEGE**

**(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)**

**Approved by AICTE &Accredited by NBA**

**KANURU, VIJAYAWADA-520007**


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# V.R.SIDDHARTHA ENGINEERING COLLEGE

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## CERTIFICATE

This is to certify that this project report titled **"MEDICAL DIAGNOSIS CHATBOT BASED ON LARGE LANGUAGE MODELS"** is a bonafide record of work done by **Hanisha Bondada (228W1A1272) , Teja Gulivindhala (228W1A1289)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

| | |
|---|---|
| **(Dr. M. Suneetha)** | **(Dr. M. Suneetha)** |
| Dean R&D, Professor &Head | Dean R&D, Professor &Head |
| Dept. of Information Technology | Dept. of Information Technology |

**EXTERNAL EXAMINER SIGNATURE**

*Date of examination:*

# DEPARTMENT OF INFORMATION TECHNOLOGY

# V.R.SIDDHARTHA ENGINEERING COLLEGE

## PROJECT SUMMARY

| S.No | Item | Description |
|---|---|---|
| 1 | Project Title | Medical Diagnosis Chatbot Based On Large Language Models |
| 2 | Student Names & Numbers | Hanisha Bondada(228W1A1272)<br>Teja Gulivindhala(228W1A1289) |
| 3 | Name of The Guide | Dr. M. Suneetha, Dean R&D, Professor &Head |
| 4 | Research Group | Artificial Intelligence in Medical Sector |
| 5 | Application Area | Health Care |
| 6 | Aim of the Project | To develop a chatbot for accurate and timely medical diagnosis support. |
| 7 | Project Outcomes | 1. Users receive instant, AI-generated medical advice based on symptoms.<br>2. Healthcare providers gain support through automated symptom analysis for consultations. |

**Student Signatures**

1.

2.

**Signature of the Guide**

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# ABSTRACT

The rise in demand for timely and accessible healthcare has motivated the creation of an AI-supported restorative conclusion chatbot, ”Ella,” from large language models such as Mistral 7B, GPT-3, PMC_LLAMA-7B, Meditron3_Gemma2-2B and Medalpaca-7b. The chatbot addresses prolonged holding up times, and restricted access in distant communities through instant, trusted, and tailored restorative answers. Prepared on a tremendous dataset of over 27 million therapeutic Q&A intelligent, the framework utilizes Characteristic Dialect Handling strategies to get it client side effects and give context-aware answers. Mistral 7B, an open-source show, was fine-tuned utilizing instruction based prompts, whereas GPT-3 served as a benchmark through API integration. Assessment measurements like BLEU and ROUGE uncovered that Mistral 7B accomplished higher precision and superior relevant pertinence than GPT-3, PMC LLAMA 7B, Meditron3-Gemma2-2B and Medalpaca-7b, in spite of the fact that with marginally expanded idleness. The chatbot is built with a Python and Node.js backend and a React.js frontend, advertising highlights such as live chat, discourse input, and inquiry history. It too incorporates usefulness to direct clients back to health-related themes when off-topic questions are inquired. The framework guarantees that clients get instructive however non-diagnostic coun sel, empowering proficient interview when fundamental. This chatbot talks to a basic step toward joining AI into healthcare back systems. Future work includes real-time information integration, sending, and investigating more progressed LLMs to assist improve symptomatic execution and openness.

**KEYWORDS:** Therapeutic Conclusion Chatbot, Indication Investigation, Mistral 7B, GPT-3, PMC LLAMA 7B, Meditron3-Gemma2-2B and Medalpaca-7b.

# CHAPTER-1

# INTRODUCTION

Healthcare is a fundamental need, but many individuals struggle to access timely and reliable medical advice. With advancements in Artificial Intelligence, especially in Natural Language Processing, AI-powered chatbots offer promising solutions. This chapter introduces the motivation, background, and problem addressed by our medical diagnosis chatbot project. It also discusses its societal relevance and provides a foundation for the work that follows.

## 1.1 ORIGIN OF PROBLEM

Healthcare accessibility is a major concern in both developing and developed countries. Patients in rural or remote areas often experience delays in receiving medical attention due to lack of transportation, inadequate healthcare infrastructure, and a shortage of professionals. Even urban healthcare systems struggle with high patient volumes, leading to long waiting times and overburdened doctors. In search of quicker solutions, many people turn to online platforms for medical information, but these sources often offer generic or inaccurate content, which can lead to misdiagnosis or delayed treatment [1].

The advancement of Artificial Intelligence (AI), especially in Natural Language Processing (NLP), provides a promising alternative. Large Language Models (LLMs) like Mistral 7B and GPT-3 are capable of understanding and generating human-like text, making them suitable for simulating intelligent medical conversations [2]. This project aims to harness the potential of these models to build a reliable, responsive medical chatbot that assists users in interpreting symptoms and making informed health decisions, especially when immediate access to a healthcare professional is not available[3].

## 1.2 BASIC DEFINITIONS AND BACKGROUND

### 1.2.1 Large Language Models (LLMs)

Large Language Models (LLMs) are advanced artificial intelligence models trained on massive amounts of textual data to understand, process, and generate human language. They are designed using deep learning techniques, particularly Transformer architectures, which allow them to handle complex language tasks such as text generation, translation, summarization, and question-answering. In the healthcare domain, LLMs can simulate human-like

conversations and offer relevant information based on the user's input, making them highly suitable for chatbot applications.

### 1.2.2 Mistral 7B

Mistral 7B is an open-weight large language model developed for high efficiency and performance. It contains 7 billion parameters and is known for its fast inference and strong accuracy in language tasks. Due to its open-source nature, it can be fine-tuned on custom datasets for specific domains, such as medical Q&A systems. In this project, Mistral 7B is fine-tuned on a medical dialogue dataset to deliver accurate, context-aware responses for symptom-based queries.

### 1.2.3 GPT-3 (Generative Pretrained Transformer 3)

GPT-3, developed by OpenAI, is one of the most powerful and widely used large language models. It consists of 175 billion parameters and is capable of generating fluent, coherent, and contextually relevant text based on prompts. GPT-3 uses a transformer-based architecture and is trained on a broad range of internet text. In this project, GPT-3 is used via API to provide instruction-based medical responses and serve as a baseline for performance comparison with Mistral 7B.

### 1.2.4 PMC LLAMA 7B

Stanford's research team created a medical language model with 7 billion parameters, training it on medical and scientific papers. They used the LLaMA system as the foundation and found it works really well for medical-related tasks, especially when doctors need answers to questions or want to summarize medical information.

### 1.2.5 Meditron3-Gemma2-2B

There's also this smaller model that Google developed. It only has 2 billion parameters but is specifically made for medical stuff. They trained it using medical research papers, doctors' notes, and recorded conversations between 4 doctors and patients. It's particularly useful when you don't have much computing power or want to use it on smaller devices.

### 1.2.6 Medalpaca-7B

It has 7 billion parameters and was specially trained on medical information using something called the Alpaca method. They made it to help answer patients' questions, assist doctors in making decisions, and create educational content about healthcare. It's good at following instructions while also knowing a lot about medical topics.

## 1.3 PROBLEM STATEMENT

In the current healthcare landscape, obtaining immediate and accurate medical advice remains a challenge. The reliance on traditional consultations is not always feasible due to financial, geographical, and temporal constraints. Moreover, many individuals lack the medical literacy needed to interpret symptoms correctly or to determine the urgency of their condition. Existing digital solutions are either too general or unreliable. This project aims to bridge this gap by developing an AI-powered medical chatbot capable of delivering context-aware, symptom-based preliminary medical guidance using fine-tuned LLMs. The chatbot is designed to function as an initial point of contact, assisting users in understanding their health condition while ensuring that professional consultation is encouraged for final diagnosis and treatment. To address these issues, the proposed solution leverages advanced language models fine-tuned on medical data to ensure accurate and personalized responses. The chatbot will analyze user-reported symptoms in real time, offering contextually relevant guidance. It aims to empower users with better health awareness and prompt action. Ultimately, the system serves as a supportive tool, not a replacement for medical professionals.

### 1.3.1 OBJECTIVES

Develop an AI-powered medical chatbot using Natural Language Processing (NLP)and a fine-tuned medical model for preliminary diagnosis.

To process and understand user input through natural language and provide accurate, context-aware responses.

### 1.3.2 OUTCOMES

Users can receive instant preliminary medical advice based on symptoms through an interactive chatbot.

Healthcare providers can use the chatbot to support patient consultations by offering AI-driven symptom analysis

## 1.4 APPLICATIONS

**Symptom-Based Preliminary Diagnosis** – Offers instant health advice based on user-described symptoms.

**Remote Healthcare Access** – Supports users in underserved or rural areas with limited medical facilities.

**Health Awareness & Education** – Educates users on common illnesses, prevention, and when to seek medical help.

## 1.5 SUMMARY

This chapter introduced the motivation and background behind developing an AI-powered medical chatbot using advanced large language models like Mistral 7B and GPT-3. It identified the core challenges in accessing reliable healthcare, particularly in remote and underserved regions.

# CHAPTER-2

# REVIEW OF LITERATURE

This chapter describes the review of literature that we have taken from various papers and considered all the points mentioned in the papers.

## 2.1 DESCRIPTION OF EXISTING SYSTEM

**1. Medical Chatbot usingLLAMA2**

**Year:**2025

**Authors:** Shaik Abdul Rahman, Mohammed Zubair, Naveed Akhter, Mohd Salman Uzair.

**Observations:**

This research presents a LLAMA2-based chatbot aimed at preliminary diagnosis and symptom analysis. It integrates LangChain, OpenAI API, and external medical databases like WebMD for real-time data. Fine-tuned on medical datasets, it achieves 92% accuracy, offering real-time, scalable, and HIPAA-compliant healthcare solutions [1].

**2.D-Care: A Multi-Tone LLM-Based Chatbot Assistant for Diabetes Patients**

**Year:**2025

**Authors:** Awais Khan Nawabi, Janos Tolgyesi, Elena Bianchi, Chiara Toffanin, Piercarlo Dondi

**Observations:**

This study introduces D-Care, a chatbot based on OpenAI's GPT-4o, designed to support diabetes patients. It uses a multi-tone approach where the chatbot responds in four different tones (elementary to technical) to suit different user needs. Retrieval-Augmented Generation (RAG) is used to ensure answer quality, and a user study with 40 participants confirmed that tone affects comprehension. The system enhances patient education and engagement, but emphasizes it should supplement, not replace, professional medical advice [2].

**3. The future of AI clinicians: assessing the modern standard of chatbots and their approach to diagnostic uncertainty**

**Year:**2024

**Authors:** Ryan S. Huang et al.

**observations:**

This research evaluates the performance of GPT-4o and Claude-3 in handling uncertain diagnostic cases, comparing them with medical residents. The study finds that AI chatbots perform well with structured medical knowledge but struggle with clinical ambiguity. These systems lack the ability to reason like experienced human doctors. While AI can offer suggestions quickly, its reliability diminishes with less straightforward cases. The authors highlight the importance of integrating human oversight in medical AI usage. The study underlines that AI tools are supplementary rather than replacements. It promotes a cautious yet collaborative approach to AI in healthcare [3].

**4. AI-Based Medical Chatbot for Disease Prediction**

**Year:** 2024

**Authors:** Ashish Zagade et al.

**observations:**

This paper presents a chatbot that uses NLP and symptom-based inputs to predict diseases effectively. The system employs JSON-formatted data structures to improve model training and enhance accuracy. It is especially valuable during emergencies like pandemics when access to care is limited. The chatbot delivers personalized feedback and supports early detection. It helps users understand symptoms and seek timely care. By leveraging AI for triage and advice, the system supports decentralized healthcare. This approach promotes proactive medical attention and better resource allocation [4].

**5. Medical Chatbot: AI Based Infectious Disease Prediction Model**

**Year:** 2024

**Authors:** Srujana Biradar & Swaroopa Shastri

**observations:**

The study focuses on a chatbot designed for predicting infectious diseases using basic ML algorithms. It features a user-friendly interface and interacts naturally with patients. The model emphasizes early detection and health guidance, especially for contagious illnesses. It aims to minimize the spread by providing timely recommendations. Simple classifiers make it

lightweight and suitable for wider deployment. The system underlines the importance of awareness in managing public health risks. Overall, it demonstrates how AI can assist in early disease management affordably [5].

**6. Conversational AI for Healthcare to Improve Member Efficiency**

**Year:** 2024

**Authors:** Nivedhitha D P, Madhumitha G, Janani Sri J, Jayashree S, Surya J, and Divya D

**observations:**

This study presents a chatbot-based healthcare assistant using the RASA Framework, Natural Language Processing (NLP), Machine Learning (ML), and the DIET classifier to optimize healthcare service delivery. The system facilitates appointment booking, real-time monitoring, medicine orders, and emergency support, even integrating Metaverse-based virtual consultations. Designed to ease administrative burdens and improve healthcare accessibility, it automates routine processes and supports multilingual interactions [6].

**7. Intelligent Chatbot for HospitalRecommendation System**

**Year:** 2024

**Authors:** Rajkumar K, Ragupathi T, and Karthikeyan S

**observations:**

This research showcases a chatbot that recommends hospitals based on user needs using ANN and NLP techniques. It functions in both online and offline environments, ensuring wide usability. The chatbot responds in real time, guiding users toward suitable healthcare facilities. It can scale to handle many users simultaneously without performance drops. The system improves the efficiency of service delivery by matching users to the right providers. It also addresses user-specific conditions to ensure relevant recommendations. This model supports both urban and rural healthcare navigation [7].

8. **Machine Learning and Deep Learning-Based Approach in Smart Healthcare: Recent Advances, Applications, Challenges and Opportunities**

**Year:** 2024

**Authors:** Anichur Rahman, Tanoy Debnath, Dipanjali Kundu

**observations:**

This survey reviews how ML and DL are transforming healthcare across multiple areas like diagnosis, imaging, and wearable tech. It explores various architectures including CNNs, autoencoders, GANs, and reinforcement learning. These models are applied in medical record analysis, smart devices, and imaging diagnostics. The paper highlights ongoing challenges like data variability and real-time processing needs. It stresses the demand for robust, adaptable, and interpretable models. The study also discusses how smart healthcare is shifting towards AI-centric solutions. Overall, it provides a comprehensive view of emerging intelligent healthcare systems [8].

9. **A chatbot-based question and answer system for auxiliary diagnosis of chronic diseases**

**Year:** 2024

**Authors:** Sainan Zhang & Jisung Song

**observations:**

This study introduces Chat Ella, a GPT-2-based chatbot tailored to diagnose 24 chronic diseases using user-provided symptoms. The model has been fine-tuned to improve its reliability in detecting commonly occurring chronic conditions. Its dialogue interface enables an intuitive user experience, delivering medical advice remotely. Achieving 97.5% accuracy, it shows strong potential for dependable pre-diagnosis. The chatbot serves as a disease management tool that reduces dependency on in-person consultations. This system aims to lessen the workload of healthcare professionals by automating chronic condition assessments. Overall, it enhances accessibility and contributes to more efficient healthcare delivery [9].

10. **Can AI Relate: Testing Large Language Model Response for Mental Health Support**

**Year:**2024

**Authors**: Saadia Gabriel, Isha Puri, Xuhai Xu, Matteo Malgaroli, Marzyeh Ghassemi

**Observations:**

This study benchmarks the equity and quality of mental health responses generated by Large

Language Models (LLMs) already deployed in clinical settings. It does not endorse LLMs for psychotherapy but critically assesses their ability to provide mental health support. The paper highlights potential concerns around fairness, reliability, and the appropriateness of LLM-generated responses in sensitive healthcare domains [10].

**11. Can ChatGPT provide intelligent diagnoses? A comparative study between predictive models and ChatGPT**

**Year**: 2024

**Authors**: Loredana Caruccio et al.

**observations:**

This paper compares ChatGPT with traditional machine learning models and Google BARD for diagnosing low- to medium-risk diseases. It emphasizes the significance of prompt engineering in improving ChatGPT's performance. The chatbot shows strengths in understanding user context and generating relevant medical responses. However, its accuracy fluctuates depending on the disease complexity. A hybrid bot is proposed to combine ChatGPT's language ability with structured models for consistency. The paper suggests that ChatGPT requires further fine-tuning for medical reliability. It concludes that while ChatGPT is promising, careful implementation is essential for clinical use [11].

**12. AI Healthcare Chatbot System using LLAMA3**

**Year:**2024

**Authors:** Prabanand S C, Akilan S, Bhoopesh S J, Ananda Kumar A, Saswin.

**observations:**

This study proposes a healthcare chatbot built with LLAMA3, capable of handling real-time medical inquiries and symptom checks. The system emphasizes personalized care, integration with EMRs, and compiance with privacy laws like HIPAA. It uses fine-tuned LLM models to provide secure, scalable healthcare support, aiming to reduce the workload on healthcare providers[12].

## 2.2 SUMMARY OF LITERATURE STUDY

1. **Medical Chatbot using LLAMA2**

   This research introduces a medical chatbot built on the LLAMA2 model, integrating NLP, LangChain, and external medical databases. Fine-tuned on medical datasets, the chatbot achieves 92% accuracy in symptom analysis and preliminary diagnosis. It offers real-time, HIPAA-compliant responses, enhancing healthcare accessibility and reducing the load on healthcare professionals. The system is designed for scalable, secure telemedicine applications [1].

2. **D-Care: A Multi-Tone LLM-Based Chatbot Assistant for Diabetes Patients**

   This paper presents D-Care, a GPT-4o-based chatbot designed to assist diabetes patients through multi-tone responses, from simple to technical language. Using Retrieval-Augmented Generation (RAG) and LangChain, the chatbot adapts its communication style based on user needs. A user study confirms that tone influences comprehension and usability. The chatbot aims to educate patients while reinforcing the importance of professional medical advice [2].

3. **The Future of AI Clinicians: Assessing the Modern Standard of Chatbots and Their Approach to Diagnostic Uncertainty**

   This research evaluates GPT-4o and Claude-3 against medical residents in uncertain diagnosis scenarios. AI performs well with structured data but struggles with ambiguous clinical reasoning. Human-like reasoning remains a challenge for current models. The study recommends pairing AI tools with healthcare professionals. It views AI as supportive, not a replacement, in clinical environments [3].

4. **AI-Based Medical Chatbot for Disease Prediction**

   This paper presents a chatbot using NLP and symptom-based inputs to predict diseases, especially useful during pandemics. JSON data improves model learning and personalization. It offers early detection and advice, promoting timely intervention. The system aims to decentralize care and improve patient awareness. It enhances access in health emergencies through AI-driven interactions [4].

5. **Medical Chatbot: AI-Based Infectious Disease Prediction Model**

   The model is designed to detect infectious diseases using basic ML classifiers in a

conversational interface. It focuses on early intervention and guidance to prevent disease spread. The chatbot is lightweight and scalable for broad deployment. It supports awareness and health management during outbreaks. The system shows how simple AI tools can assist public health [5].

**6. Conversational AI for Healthcare to Improve Member Efficiency**

This paper explores a healthcare assistant chatbot built on the RASA framework with DIET classifiers and NLP. It handles tasks like bookings, medication tracking, and virtual consultations. Integration with the Metaverse is also supported. The model improves healthcare automation and real-time support. It enhances service delivery in both remote and urban regions [6].

**7. Intelligent Chatbot for Hospital Recommendation System**

The research describes a chatbot using ANN and NLP to recommend hospitals based on user needs. It works in offline and online modes for flexibility. The chatbot offers real-time responses and scales for high user traffic. It matches users with appropriate facilities based on symptoms. This improves hospital accessibility and efficiency across regions [7].

**8. Machine Learning and Deep Learning-Based Approach in Smart Healthcare**

This survey reviews ML/DL use in diagnosis, medical imaging, EHR analysis, and wearable devices. It explores CNNs, GANs, and reinforcement learning in smart healthcare. The study highlights issues like data heterogeneity and the need for real-time decisions. It promotes the shift toward AI-based, interpretable health solutions. Overall, it offers a comprehensive view of intelligent healthcare technologies [8].

**9. A Chatbot-Based Question and Answer System for Auxiliary Diagnosis of Chronic Diseases**

This study introduces Chat Ella, a GPT-2-based chatbot fine-tuned for diagnosing 24 chronic diseases. It delivers 97.5% accuracy using symptom-based dialogue, providing users with remote diagnostic support. The system enhances healthcare accessibility and reduces dependency on physical consultations. The model is particularly effective in chronic illness pre-diagnosis and patient engagement [9].

**10. Can AI Relate: Testing Large Language Model Response for Mental Health Support**

This study benchmarks the quality and fairness of LLMs used in mental health support settings.

It critically assesses LLMs like GPT models without endorsing them for psychotherapy. The research highlights the need for cautious deployment, focusing on equity, quality of care, and the limitations of AI in sensitive mental health interactions. The paper emphasizes that LLMs can assist but should not replace trained professionals [10].

**11. Can ChatGPT Provide Intelligent Diagnoses? A Comparative Study Between Predictive Models and ChatGPT**

This paper compares ChatGPT, traditional models, and Google BARD for diagnosing low- to mid-risk diseases. It emphasizes prompt engineering and proposes a hybrid bot for improved support. While ChatGPT excels in contextual understanding, its accuracy varies with disease complexity. The study suggests ChatGPT needs fine-tuning for clinical reliability. It concludes that ChatGPT shows potential, but human oversight is essential [11].

**12. AI Healthcare Chatbot System using LLAMA3**

This study proposes a healthcare chatbot using the LLAMA3 model for real-time symptom checking and health advice. It supports EMR integration and complies with HIPAA standards for data security. Designed for personalized patient engagement, it handles routine queries to free up medical staff for critical tasks. The system emphasizes scalability, accuracy, and patient safety in digital healthcare environments [12].

The Detailed Literature Review is shown in Table. 2.2

***Table. 2.2 Literature Review***

| S No | Title and Publication Details | Algorithms/ Technologies used | Dataset | Purpose | Remarks |
|---|---|---|---|---|---|
| 1 | Year:2025 Medical Chatbot using LLAMA2 | LLAMA2, LangChain, OpenAI API, | Public medical datasets | Symptom analysis, preliminary diagnosis, real-time healthcare support | Achieved good accuracy, HIPAA-compliant |

| | | | | | |
|---|---|---|---|---|---|
| 2 | Year:2025<br>D-Care: A Multi-Tone LLM-Based Chatbot Assistant for Diabetes Patients | OpenAI GPT-4o,RAG LangChain | Diabetes-related medical corpus | Improve diabetes patient education with multi-tone chatbot | Tone affects comprehension and usability |
| 3 | Year:2024<br>Title: The future of AI clinicians: assessing the modern standard of chatbots and their approach to diagnostic uncertainty | GPT-4o, Claude-3 | Clinical cases curated from hospital records and diagnostic case banks | Analyze AI in uncertain diagnoses | AI promising but lacks clinical reasoning |
| 4 | Year:2024<br>Title: AI-Based Medical Chatbot for Disease Prediction | NLP, symptom-based input | Synthetic symptom dataset generated for common diseases | Predict diseases using chatbot interface | Enhances access, pandemic-relevant tool |
| 5 | Year:2024<br>Title: Medical Chatbot: AI Based Infectious Disease Prediction Model | NLP, ML classifiers | infectious disease symptoms dataset from WHO and CDC reports | Early detection of infectious diseases | Simple interface, early prediction focus |
| 6 | Year:2024<br>Title: Conversational AI for Healthcare to Improve Member Efficiency | NLP, DIET classifier | Custom- built healthcare management dataset including patient interactions. | Automate services like booking, emergency help | Supports multilingual and virtual care |

| | | | | | |
|---|---|---|---|---|---|
| 7 | Year:2024<br>Title: Intelligent Chatbot for Hospital Recommendation System | NLP, ANN | Healthcare queries and hospital info from medical service platforms | Recommend hospitals and handle queries | Dual-mode system, scalable design |
| 8 | Year:2024<br>Title: Machine Learning and Deep Learning-Based Approach in Smart Healthcare | CNNs,Auto-encoders, GANs, RL | :Uses multiple sources like EHRs, medical images. | Review ML/DL in healthcare domains | Summarizes advances, highlights challenges |
| 9. | Year: 2024<br>Title: A chatbot-based question and answer system for auxiliary diagnosis of chronic diseases. | GPT-2,fine-tuning | Symptom dataset from open health databases. | Diagnose 24 chronic diseases and provide advice. | Latency issue, chronic disease focus. |
| 10. | Year:2024<br>Can AI Relate: Testing Large Language Model Response for Mental Health Support | OpenAI GPT- 4o, RAG,Lang Chain. | Diabetes-related medical corpus | Improve diabetes patient education with multi-tone chatbot | Tone affects comprehension and usability |

| | | | | | |
|---|---|---|---|---|---|
| 11. | year:2024<br>Title: Can ChatGPT provide intelligent diagnoses?A comparative study between predictive models and ChatGPT | ChatGPT, Google BARD, ML models | Public health symptom datasets covering low- to medium-risk diseases | Compare diagnostic ability of LLMs vs ML | ChatGPT shows context strength but variable accuracy |
| 12. | Year:2024<br>AI Healthcare Chatbot System using LLAMA3 | LLAMA3, NLP,EMR Integration, Cloud deployment | Public and clinical medical datasets | Personalized healthcare support, symptom checking | Focuses on secure, scalable AI healthcare integration |

Existing medical chatbots struggle with diagnosing complex or uncertain cases accurately. Many lightweight models are fast but lack deep medical understanding. Larger models demand heavy computational resources and time. Some systems fail to reason like experienced doctors, especially in ambiguous scenarios.

The proposed system using Mistral 7B and GPT-3 ,PMC LLAMA 7B, Meditron3-Gemma2-2B and Medalpaca-7b, enhances deep reasoning and medical context understanding.. These models offer high scalability without sacrificing performance. They reduce dependency on heavy computing by using optimized architectures. Ultimately, the system ensures reliable and intelligent healthcare support.

## 2.3 SOFTWARE REQUIREMENT SPECIFICATION

### 2.3.1 Purpose:

The purpose of this project is to design and develop an AI-powered medical diagnosis chatbot capable of providing preliminary medical assistance using fine-tuned Large Language Models (LLMs) specifically Mistral 7B ,GPT-3, PMC_LLAMA_7B, Meditron3-Gemma2-2B and Medalpaca-7b. The chatbot should be able to understand patient symptoms, match them with medical conditions, and offer accurate, context-aware, and reliable responses. It aims to improve accessibility, reduce response time, and support healthcare professionals in delivering virtual

consultation support.

### 2.3.2 Scope:

This project aims to build an AI-based medical chatbot that provides instant, preliminary health advice using Mistral 7B,GPT-3, PMC_LLAMA_7B, Meditron3-Gemma2-2B and Medalpaca-7b language models. Users interact through a React.js interface, while backend processing and model inference are handled via Python and Node.js. The chatbot analyzes symptoms and delivers context-aware responses, storing interactions in a PostgreSQL database. It is designed to improve healthcare accessibility and may be extended with features like voice input, multilingual support, and cloud deployment.

### 2.3.3 Intended Users:

- Patients seeking quick health advice
- Doctors and healthcare workers assisting diagnosis
- Medical researchers evaluating AI models
- Health tech companies building smart healthcare tools

### 2.3.4 Functional Requirements:

- The system should accept text-based symptom inputs from users via a React.js interface.
- It should support real-time chat interaction and maintain conversation history.
- The chatbot should process input and route it to Mistral 7B or GPT-3 or PMC LLAMA 7B, Meditron3 Gemma2-2B , Medalpaca-7b based on configuration.
- All user queries and chatbot responses should be stored in a PostgreSQL database.

### 2.3.5 Non-Functional Requirements:

- Basic security measures such as input validation, session control, and data protection must be implemented.
- The application should maintain an uptime of at least 99% during normal usage.
- The user interface must be intuitive, responsive, and accessible.
- The chatbot must respond within 2 seconds to maintain a smooth user experience.

### 2.3.6 Software Requirements:

**Backend:**

Language: Python (for model logic), Node.js (API handling)

Libraries: transformers, torch, langchain, openai, sklearn, huggingface_hub

Server: Express (Node.js), Flask or FastAPI (optional for Python endpoints)

Database: PostgreSQL 13+

**Frontend:**

Framework: React.js

Libraries: axios, socket.io-client, styled-components, react-icons

Model and Infrastructure:

LLMs: Mistral 7B (local), GPT-3 (OpenAI API)

Fine-tuning Platform: Google Colab / Hugging Face CLI / OpenAI CLI

## 2.4 Summary

This chapter reviews recent research on AI-based medical chatbots for disease prediction and healthcare support. It highlights their effectiveness, usability, and the importance of human oversight in clinical applications.

# CHAPTER-3

# PROPOSED METHODOLOGY

This chapter summarizes the design methodology, system architecture, core modules, algorithms, datasets, and tools used for the development of the chatbot for medical diagnosisusing Large Language Models (LLMs) such as Mistral 7B and GPT-3 PMC LLAMA 7B, Meditron3-Gemma2-2B and Medalpaca-7b.

## 3.1 DESIGN METHODOLOGY

### 1 Data Collection and Preprocessing

The chatbot is trained on a publicly available dataset from Kaggle titled "AI Medical Chatbot", consisting of over 242,000 doctor-patient interactions. The dataset includes both patient queries and medical responses, which are cleaned and preprocessed to remove noise such as unnecessary symbols, inconsistent casing, and irrelevant entries. Tokenization is applied using appropriate tokenizers from Hugging Face for Mistral and GPT-3 PMC LLAMA 7B, Meditron3-Gemma2-2B and Medalpaca-7b models. Instruction-style formatting is used to convert question-answer pairs into prompt-response templates suitable for fine-tuning.

### 2 Model Loading and Fine-Tuning

Mistral: 7B: The pretrained Mistral 7B model is loaded using Hugging Face's transformers library with AutoModelForCausalLM and AutoTokenizer. It is fine-tuned in Google Colab using a medical Q&A dataset and optimized with instruction-based prompts.

GPT-3: GPT-3 is accessed via the OpenAI API and fine-tuned using structured JSONL data through the OpenAI CLI. It requires no local setup, as it runs on OpenAI's cloud infrastructure with secure API access.

PMC_LLaMA-7B: A LLaMA-7B base model fine-tuned on PubMed Central (PMC) papers (S2ORC) to improve medical/scientific text generation and QA. It's distributed on Hugging Face and typically loaded with the transformers API like other LLaMA variants.

Meditron3-Gemma2-2B: A research-oriented, medicine-specialized LLM that builds on Gemma-2 (2B) architecture for clinical and low-resource healthcare use cases; intended for evidence-based medical assistance and evaluation in research settings. It's available from the Meditron project on Hugging Face and usually used for experiment/benchmarks rather than clinical deployment.

MedAlpaca-7B: A 7-billion-parameter LLaMA derivative fine-tuned on curated medical Q&A and dialogue datasets (MedAlpaca family) to improve medical question-answering and conversational performance; commonly loaded via transformers and used for open-source medical chatbot development.

**3 Evaluation During Training**

To effectively monitor and compare the performance of both Mistral 7B and GPT-3 during training, multiple evaluation metrics are utilized. BLEU and ROUGE scores are employed to assess the linguistic quality and relevance of the generated responses compared to ground truth answers. Latency is also recorded to evaluate the response time of each model, which is a critical factor for real-time chatbot interaction. In addition, confusion matrices and scorecards are generated to analyze the accuracy and contextual reliability of the responses. Both models are tested on a dedicated validation set consisting of unseen medical questions to ensure their generalizability and robustness.

**4 Prediction and User Interaction**

After fine-tuning, the chatbot is capable of processing real-time user inputs for medical diagnosis. When a user submits a query through the React.js interface, the input text is tokenized and routed to the selected model either Mistral 7B or GPT-3 based on system configuration. The model then generates a contextually relevant medical response, which is formatted and displayed to the user. All interactions, including queries, model outputs, and timestamps, are logged in a PostgreSQL database for future analysis and optimization. The chatbot also includes logic to identify and handle non-medical or irrelevant queries, gently redirecting users to maintain focus on healthcare-related interactions.

**5 Frontend Development**

User Interface Considerations: Mention the thought process behind designing the React.js interface, including user experience decisions such as accessibility, user input validation, and response display.

API Integration Details: You can also provide more context about how the backend (Node.js + Python) and frontend (React.js) communicate, including the use of APIs for smooth interaction between the chatbot models and the user interface.

## 3.2 SYSTEM ARCHITECTURE

The system architecture of the medical diagnosis chatbot integrates GPT-3 with a structured

backend and intuitive frontend to deliver instant, AI-driven medical responses.

1 **Dataset**

The project utilizes the AI Medical Chatbot dataset sourced from Kaggle, consisting of over 242,000 real medical interactions between patients and doctors. This dataset serves as the foundation for training and evaluating the GPT-3 model. It covers a broad range of symptoms, conditions, treatments, and diagnostic discussions, offering rich domain-specific context. The data is carefully cleaned to remove inconsistencies and formatted to support both keyword-based matching and NLP-based fine-tuning workflows. Special attention is given to maintaining data balance across different medical specialties. Privacy of patient information is preserved during preprocessing. This preparation ensures a consistent and reliable knowledge base for the chatbot. As a result, the model is able to generate fast, accurate, and context-aware medical responses.

2 **Model Training:**

**Tokenization:** Both models use tokenization to convert text into smaller units, making it easier for the models to process inputs.

**Hyperparameters:** Hyperparameters like learning rate, batch size, and epochs were optimized to improve model performance during fine-tuning.

**Models Used:** Mistral 7B is a fine-tuned open-weight model for medical tasks, while GPT-3 serves as a baseline for comparison.

**Fine-Tuning:** Fine-tuning was done using 80% of the dataset, focusing on instruction-based prompts for medical Q&A adaptation.

3 **Evaluation of Models**

Model performance is validated using BLEU and ROUGE scores standard metrics for measuring the quality of generated text.

BLEU - Measures the precision of word overlap between the chatbot’s response and reference answers.

ROUGE - Evaluates recall by comparing the number of shared phrases between the model's output and expected answers.

LATENCY -Tracks the response time of the model to ensure quick interaction.

## 4 Chatbot Interface

The chatbot interface, built with React.js, offers a clean, user-friendly experience with text and voice input support. It features typing indicators, chat history, and options to clear or end the chat, allowing natural and responsive medical conversations. The UI ensures users stay focused on health-related topics for reliable assistance.

Example:User: "I have a headache and feel tired."

Chatbot: " hydrated with water, herbal teas, or electrolyte solutions,Rest in a cool room and use damp cloths to reduce discomfort,Monitor your temperature and symptoms closely."

System Architecture diagram is shown in Figure 3.1

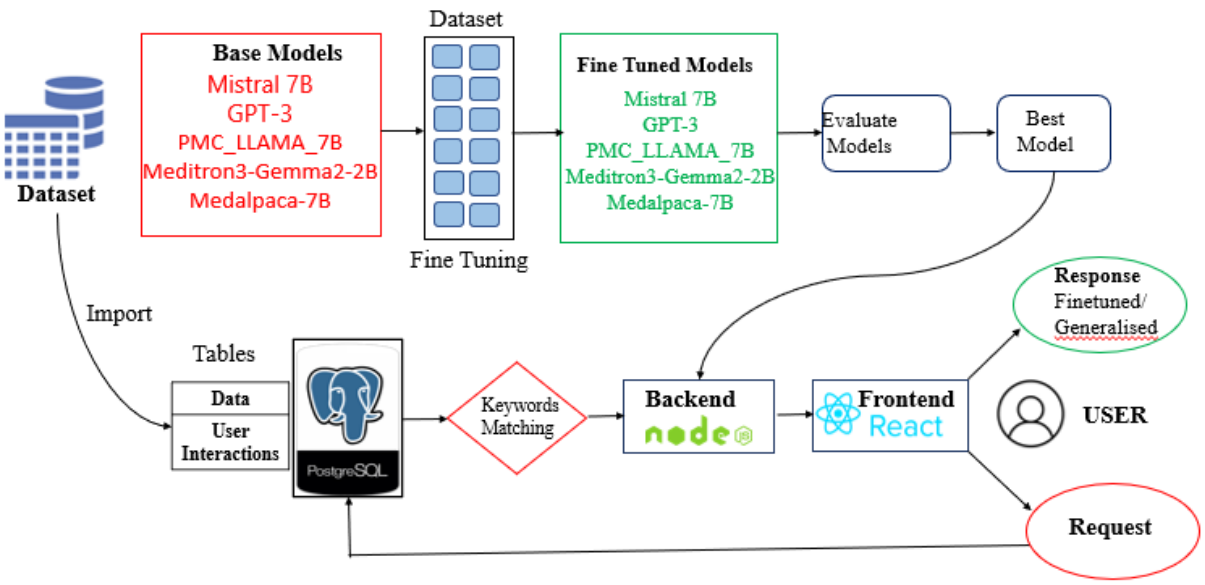


***Figure 3.1 Architecture Diagram***

The Figure 3.1 illustrates a system where a dataset is used to fine-tune base models like Mistral 7B and GPT-3, after which the models are evaluated and the best-performing one is selected. Simultaneously, the dataset is imported into a PostgreSQL database to store user data and interactions, enabling a keyword matching mechanism. A Node.js backend handles the processing and connects to a React-based frontend where users send requests and receive either fine-tuned or generalized responses. The system continuously improves by learning from new user interactions stored back in the database.

## 3.3 DESCRIPTION OF ALGORITHMS

**1 Mistral 7B**:

Mistral 7B is a state-of-the-art open-weight large language model (LLM) designed for causal language modeling. It generates coherent and contextually relevant text based on input prompts. For your medical chatbot, Mistral 7B was fine-tuned on a medical Q&A dataset, enabling it to understand and respond to medical queries effectively. The model uses a transformer architecture and is trained with supervised fine-tuning using instruction-style prompts for improved medical dialogue.

**Purpose:** Used as the primary model for the chatbot, fine-tuned to generate accurate and context-aware responses to medical queries**.**

Mistral 7B Algorithm Steps are shown in Table 3.2

***Table 3.2 Mistral 7B Algorithm***

**Algorithm:** Mistral-7B for Medical Chatbot Response Generation

**Input:** Preprocessed medical Q&A dataset

**Output:** Accurate, context-aware medical chatbot responses

**Method:**

1. Load Mistral-7B model and define prompt templates.
2. Preprocess and tokenize the medical Q&A dataset.
3. Fine-tune the model with supervised learning, applying early stopping and hyperparameter tuning.
4. For each epoch:
   - Pass tokens and attention masks to the model.
   - Compute output logits and loss.
   - Backpropagate loss to update weights.
5. Evaluate the model using BLEU, ROUGE, and Latency metrics.

$$\text{BLEU}=\text{BP}\cdot\exp\left(\sum_{n=1}^{N} W_n . \log p_n\right) \quad \text{[Eq-1]}$$

$$\text{ROUGE}=\frac{Longest\ \ Common\ \ Subsequence}{Lenght\ of\ Reference\ Text} \quad \text{[Eq-2]}$$

$$\text{LATENCY}=\text{Response Time}-\text{Input Tim} \quad \text{[Eq-3]}$$

6. Generate responses using the fine-tuned model.

**2 GPT-3:**

GPT-3, developed by OpenAI, is another powerful LLM used in your project. It utilizes a transformer-based architecture and excels in understanding and generating human-like text. The model is accessed via the OpenAI API, enabling it to handle complex and varied language inputs, including medical queries. GPT-3 was used as a baseline for comparison, with fine- tuning achieved through OpenAI's infrastructure using structured JSONL data to optimize it for medical dialogues.

**Purpose:** Used as a baseline model to compare performance and ensure response quality, especially for handling complex or ambiguous medical questions.

GPT-3 Algorithm steps are shown in Table 3.3

***Table 3.3 GPT-3 Algorithm***

**Algorithm: GPT-3 for Medical Chatbot Response Generation Input**:

Instruction-based medical queries (formatted prompts)

**Output**: Accurate, context-aware medical chatbot responses via OpenAI API

**Method**:

1. **Load GPT-3 Model**
   - Access GPT-3 via Hugging Face or OpenAI API.
   - Define structured prompt templates.
2. **Preprocess Data**
   - Clean and format the medical Q&A dataset.
3. **Fine-tune the Model**
   - Convert Q&A into JSONL format.
   - Fine-tune GPT-3 on domain-specific medical data using OpenAI CLI/API.
4. **For Each Epoch**:
   - Pass structured prompts to the model.
   - Compute output logits and loss.
   - Backpropagate loss to update model weights.
5. **Evaluate the Model**
   - Measure response quality using BLEU, ROUGE, and Latency metrics.
     - BLEU [Eq-1]
     - ROUGE [Eq-2]
     - Latency [Eq-3]
6. **Generate Responses**
   - Use the fine-tuned model to generate context-aware medical responses.

**3 PMC_LLaMA-7B:**

PMC LLaMA-7B is a specialized version of the LLaMA model that has been fine-tuned on medical research articles taken from PubMed Central (PMC). Because it is trained on large-scale biomedical literature, it develops stronger understanding of medical terms, clinical concepts, research language, and evidence-based reasoning compared to a general LLaMA model.

**Purpose:**The main purpose of PMC LLaMA is to generate accurate, research-supported medical text. It is used for medical question-answering, summarizing clinical research papers, assisting in biomedical chatbots, and supporting evidence-based decision-making by providing scientifically grounded responses.

PMC_Llama-7B Algorithm steps are shown in table 3.4

***Table 3.4 PMC_LLAMA-7B Algorithm***

**Algorithm: PMC_LLaMA-7B**

**Input:** Medical Q&A + PMC research text

**Output:** Evidence-aware medical responses

**Method:**

1. **Load PMC_LLaMA-7B Model**
   - Load via Hugging Face (LLaMA architecture).
   - Define evidence-based prompt templates.
2. **Preprocess Data**
   - Clean and format PMC/Q&A pairs.
3. **Fine-tune the Model**
   - Convert dataset to JSONL.
   - Fine-tune using supervised training or LoRA.
4. **For Each Epoch:**
   - Pass tokens + masks.
   - Compute logits and loss.
   - Backpropagate and update weights.
5. **Evaluate the Model**
   - Use BLEU, ROUGE, and Latency.
6. **Generate Responses**
   - Produce citation-friendly medical answers.

**4 Meditron3-Gemma2-2B:**

Meditron3-Gemma2-2B is a compact medical language model built on the Gemma-2B architecture and further trained on high-quality clinical and biomedical datasets. Despite its small size, it is optimized to understand medical terminology, patient symptoms, clinical notes, and diagnostic reasoning, making it efficient for low-resource environments.

**Purpose**:The primary purpose of Meditron3-Gemma2-2B is to provide safe, concise, and clinically relevant responses for medical chatbots, clinical decision-support prototypes, and healthcare research tasks. It is designed for fast inference, low computational cost, and reliable performance in medical Q&A and clinical information retrieval.

Meditron3-Gemma2-2B Algorithm steps are shown in table 3.5

***Table 3.5 Meditron3-Gemma2-2B Algorithm***

**Algorithm: Meditron3-Gemma2-2B for Medical Chatbot Response Generation**

**Input:** Clinical Q&A dataset

**Output:** Concise, clinically-oriented responses

**Method:**

1. **Load Meditron3-Gemma2-2B Model**
   - Load via Hugging Face (Gemma-2B backbone).
   - Define clinical prompt formats.
2. **Preprocess Data**
   - Clean, de-identify, and tokenize clinical text.
3. **Fine-tune the Model**
   - Convert Q&A to JSONL.
   - Fine-tune with supervised learning + early stopping.
4. **For Each Epoch:**
   - Pass tokens and masks.
   - Compute logits and loss.
   - Backpropagate to update weights.
5. **Evaluate the Model**
   - BLEU, ROUGE, and Latency metrics.
6. **Generate Responses**
   - Output short, safe clinical answers.

**5 MedAlpaca-7B:**

MedAlpaca-7B is a 7-billion-parameter medical language model built on the LLaMA architecture and fine-tuned using Alpaca-style instruction datasets combined with curated medical Q&A data. This tuning allows the model to understand medical instructions, explain health topics clearly, and engage in smooth, conversational dialogue.

**Purpose:**

The main purpose of MedAlpaca-7B is to generate user-friendly, conversational medical responses for chatbots, virtual assistants, and educational tools. It is designed to simplify complex medical information and provide helpful guidance while maintaining an easy-to-understand communication style.

MedAlpaca-7B Algorithm steps are shown in table 3.6

***Table 3.6 MedAlpaca-7B Algorithm***

**Algorithm: MedAlpaca-7B for Medical Chatbot Response Generation**

**Input:** Instruction-style medical Q&A

**Output:** Conversational medical responses

**Method:**

1. **Load MedAlpaca-7B Model**
   - Load via Hugging Face (LLaMA-Alpaca tuned).
   - Define instruction-based prompt templates.
2. **Preprocess Data**
   - Clean and convert Q&A to Alpaca-style JSONL.
3. **Fine-tune the Model**
   - Apply supervised fine-tuning or LoRA.
4. **For Each Epoch:**
   - Pass prompts/tokens.
   - Compute logits and loss.
   - Backpropagate weight updates.
5. **Evaluate the Model**
   - BLEU, ROUGE, Latency.
6. **Generate Responses**
   - Produce conversational, user-friendly medical outputs.

## 3.4 DESCRIPTION OF DATASET AND TOOLS

### 3.4.1 Dataset Description

Source: The dataset used for this project is the AI Medical Chatbot Dataset, publicly available on Kaggle and curated from diverse medical dialogues.

https://www.kaggle.com/datasets/yousefsaeedian/ai-medical-chatbot

Size and Scale:

- Patient Queries: 228,722
- Doctor Responses: 246,006
- Approximate Total Interactions: ~242,150

Structure: Each data point consists of a medical query from a patient, and one or more responses from a doctor. The dataset is designed to simulate real-world medical dialogue scenarios and includes:.

## 3.5 SUMMARY

This chapter outlines the design, architecture, and implementation of a medical chatbot using Mistral 7B ,GPT-3 ,PMC_LLaMA-7B, Meditron3-Gemma2-2B and MedAlpaca-7B, covering dataset preparation, model fine-tuning, evaluation, and user interaction. It also details the tools, algorithms, and frontend-backend integration used for real- time medical query responses.

# CHAPTER-4

# RESULTS AND OBSERVATIONS

This chapter summarizes the experimental results obtained using Mistral-7b and GPT-3 models. It presents key performance metrics, including BLEU,ROUGE,LATENCY Scores and compares their effectiveness in answering queries. Observations highlight the models' capabilities and limitations, with a focus on selecting best model.

## 4.1 STEPWISE DESCRIPTION OF RESULTS

### 1. BLEU Score (Bilingual Evaluation Understudy)

**Purpose:**

Measures how closely a machine-generated response matches a reference (human) response based on n-gram overlap.

**Formula ( for BLEU-N)** : [Eq -1]

Where:

- BP = brevity penalty

$$\text{BP}=\sum\begin{cases} 1, if c > r \\ \exp\left(1-\frac{r}{c}\right), \quad n\ even \end{cases} \qquad \text{[Eq-1]}$$

- c = length of candidate response
- r = length of reference response

**Interpretation:**

- BLEU = closer to 1(Perfect match)
- BLEU closer to 0 = less similarity

**2. ROUGE Score (Recall-Oriented Understudy for Gisting Evaluation)**

**Purpose:**

Evaluates overlap between the generated response and reference by computing recall, precision, and F1-score based on n-gram and longest common subsequence (LCS).

**Common Types:**

- **ROUGE-N:** Measures n-gram overlap
- **ROUGE-L:** Measures longest common subsequence (LCS)

**Formula (for ROUGE-N Recall)** :

$$ROUGE - N = \frac{\sum_{S \in \text{Reference}} \sum_{n-gram \in S} Count_{match}(n-gram)}{\sum_{S \in \text{Reference}} \sum_{n-gram \in S} Count(n-gram)} \quad \text{[Eq-2]}$$

**Interpretation:**

- Higher ROUGE → better coverage of important content

**3. Latency**

**Purpose:**

Measures the time taken by the model to generate a response to a user input.

**Formula( for Latency)** :

$$Latency = t_{end} - t_{start} \quad \text{[Eq-3]}$$

**Interpretation:**

- Lower latency = faster response and Target in healthcare chatbots: ≤ 2 seconds (for real-time interactions)

## 4.2 MODELS PERFORMANCE COMPARISION

Comparision of Model Performances is shown is Figure 4.1,Figure4.2,Figure4.3,Figure 4.4 .

### 4.2.1 Comparison of Mistral-7B and GPT-3

A performance comparison of Mistral-7B and GPT-3 is shown in Figure 4.1. Mistral-7B scores higher than GPT-3, with BLEU at 0.88 and ROUGE at 0.91, which means it gives more accurate and relevant answers. However, GPT-3 has a faster response time, with a latency of 1.2 seconds compared to 1.9 seconds for Mistral-7B. This comparison highlights the trade-off between output quality and efficiency: GPT-3 offers quicker, slightly less accurate results, while Mistral-7B focuses on providing precision. Comparision of Mistral-7B and GPT-3 is shown in Figure 4.1

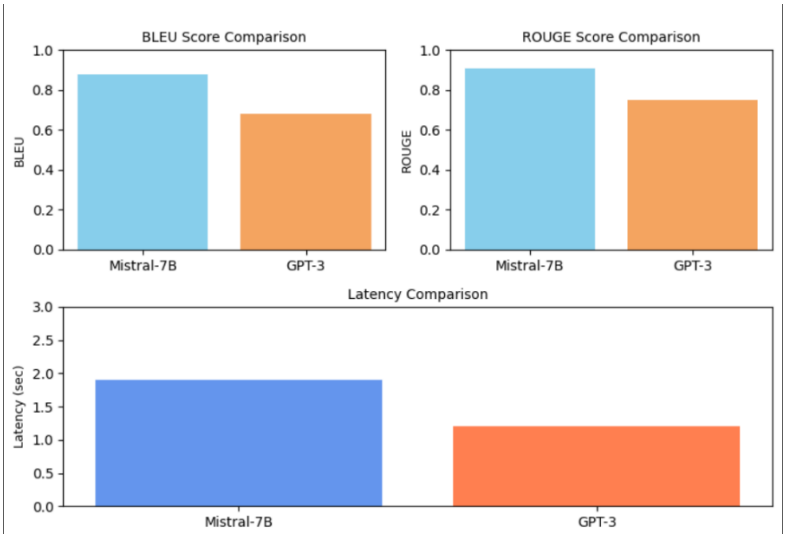


***Figure. 4.1: Comparison of Mistral-7B and GPT-3 on BLEU, ROUGE, and Latency Metrics***

### 4.2.2 Comparison of Mistral-7B and PMC LLAMA 7B

Figure 4.2 illustrates a comparison of the performance of Mistral-7B and PMC LLAMA 7BintermsofBLEU,ROUGE,andlatency.Mistral-7Bachieves better scores in BLEU and ROUGE (BLEU = 0.88, ROUGE = 0.91) than PMC LLAMA 7B (BLEU = 0.83, ROUGE = 0.87). This indicates that the responses are more relevant and accurate. PMC LLAMA 7B responds faster than Mistral-7B, whose latency is 1.5 seconds compared to 1.9 seconds. This indicates that Mistral-7B prioritizes accuracy, while PMC LLAMA 7B prioritizes speed.Comparision  Comparison of Mistral-7B and PMC LLAMA 7B is shown in Figure  4.2.

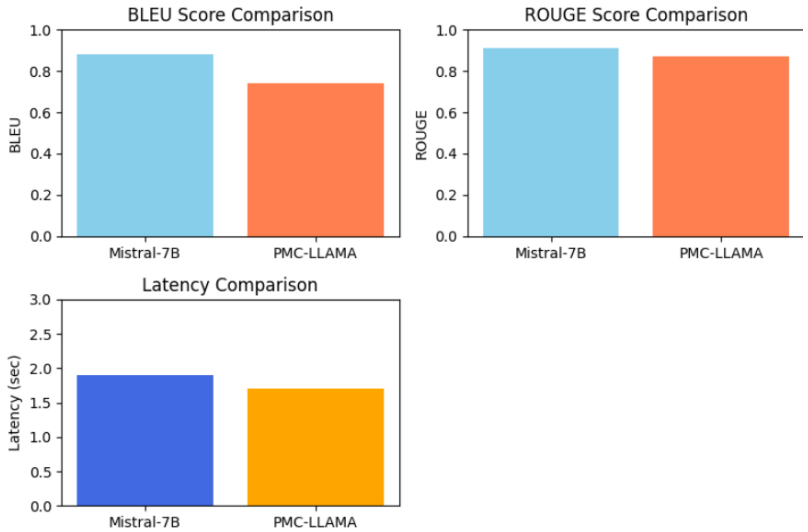


***Figure. 4.2: Comparison of Mistral-7B and PMC LLAMA 7B on BLEU, ROUGE, and Latency Metrics***

### 4.2.3 Comparison of Mistral-7B and Meditron3-Gemma2-2B

Figure 4.3 shows a graph comparing the performance of Mistral-7B and Meditron3 Gemma2-2BusingBLEU,ROUGE,andlatency.Mistral-7Bhasgreater values for both BLEU and ROUGE (BLEU = 0.88, ROUGE = 0.91) than Meditron3-Gemma2-2B (BLEU =0.81, ROUGE=0.85), which means that Mistral-7B provides more accurate and relevant responses. On the speed aspect, Meditron3-Gemma2-2B is quicker, with its latency being 1.4 seconds, while that of Mistral-7B is 1.9 seconds. It would then be safe to say that Mistral-7B is more accuracy-oriented, while Meditron3-Gemma2-2B is more concerned with quicker responses. Comparison of Mistral-7B and Meditron3-Gemma2-2B shown in Figure 4.3.

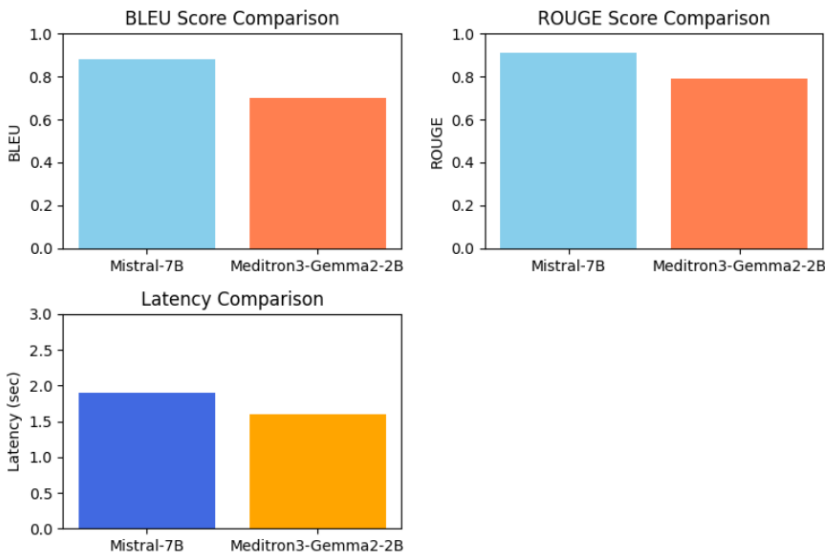


***Fig. 4.3: Comparison of Mistral-7B and Meditron3-Gemma2-2B on BLEU, ROUGE, and Latency Metrics***

### 4.2.4 Comparison of Mistral-7B and MedAlpaca-7B

Figure 4.4 shows a comparison of BLEU, ROUGE, and latency of Mistral-7B and MedAlpaca-7B. MedAlpaca-7B performs better than Mistral-7B on both BLEU and ROUGE with a BLEU of 0.92 and a ROUGE of 0.94, while that of Mistral-7B is a BLEU of 0.88 and a ROUGE of 0.91. On the issue of latency, MedAlpaca-7B is also 9 marginally quicker, responding in 1.7 seconds compared to 1.9 seconds of Mistral-7B. These data confirm that MedAlpaca-7B provides both improved accuracy and quicker responses, thus being a better fit for applications needing efficiency without affecting quality.Comparision of Mistral-7B and MedAlpaca-7B is shown in Figure 4.4.

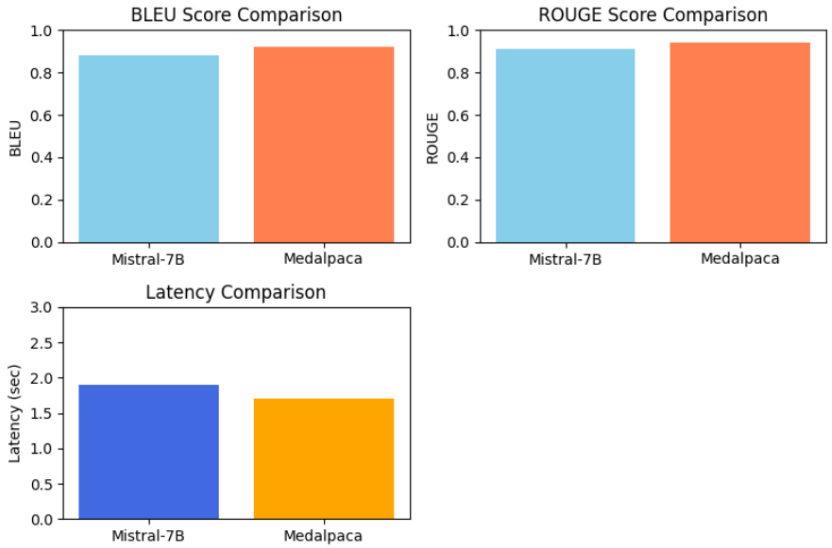


***Figure 4.4. Comparison of Mistral-7B and MedAlpaca-7B on BLEU, ROUGE, and Latency Metrics***

## 4.3 RESULT ANALYSIS

### Model summary

The system uses five models—Mistral-7B, GPT-3, PMC-LLaMA, Meditron3-Gemma2-2B, and MedAlpaca-7B—to support medical diagnosis by analyzing user queries. Their performance is evaluated using a medical Q&A dataset and measured through BLEU, ROUGE, and latency metrics. The overall comparison of all models is presented in Table 4.1.

***Table 4.1 Result Analysis for Models***

| *Model* | *BLEU Score* | *ROUGE Score* | *Latency (sec)* |
|---|---|---|---|
| ***Mistral-7B*** | ***0.88*** | ***0.91*** | ***1.9*** |
| ***GPT-3*** | ***0.680*** | ***0.750*** | ***1.2000*** |
| ***PMC-LLaMA*** | ***0.74*** | ***0.87*** | ***1.7*** |
| ***Meditron3*** | ***0.7*** | ***0.79*** | ***1.6*** |
| ***MedAlpaca-7B*** | ***0.92*** | ***0.94*** | ***1.7*** |

Table 4.1 shows that MedAlpaca-7B provides the highest accuracy, followed by Mistral-7B, while PMC-LLaMA and Meditron3 offer moderate performance and GPT-3 ranks lowest in BLEU and

ROUGE scores. In terms of speed, Meditron3 responds the fastest, with GPT-3 also performing well, whereas the remaining models have slightly higher latency. Overall, MedAlpaca-7B is best for accuracy, and Meditron3 is best for quick responses.

**Process and Results:**

**Input:** A medical query from the user, such as symptoms or health-related concerns.

"Ella" is a medical chatbot that assists users with symptoms or health concerns. It supports text and voice input, with easy options to clear, export, or end the chat is shown in Figure 4.5.

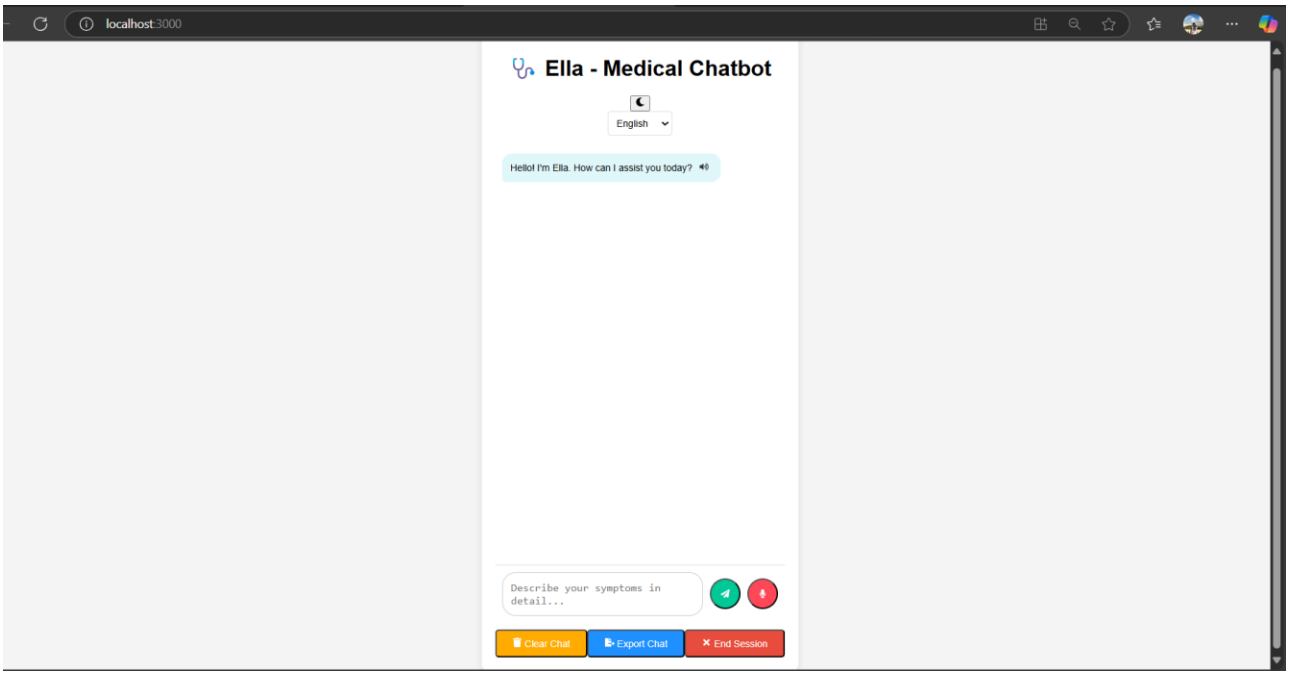


***Figure 4.5 Medical Chatbot Interface***

The Figure 4.5 represents the user interface of **Ella - Medical Chatbot**, designed to assist users in describing their symptoms and receiving medical guidance. It features a clean layout with language selection, a chat area for interaction, and options to clear the chat, export the conversation, or end the session And how it is analyzing symptoms is shown in Figure 4.6.

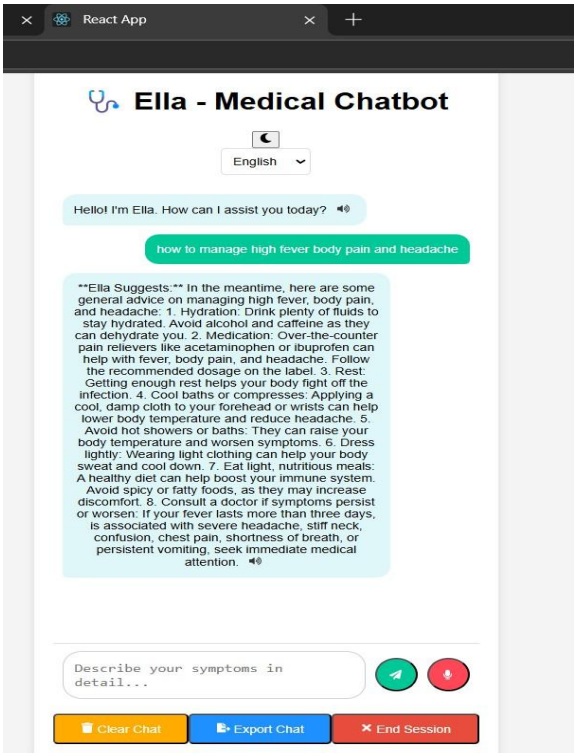


***Figure 4.6 Ella Analyzing Symptoms***

Figure 4.6 illustrates the chatbot's interaction flow, detailing each stage from receiving the user's query to generating a contextually accurate response. It shows how the chatbot first processes and understands the input, analyzes it against the trained medical knowledge base, and then formulates an appropriate and supportive reply And rejecting messages unrelated to health is shown in Figure 4.7.

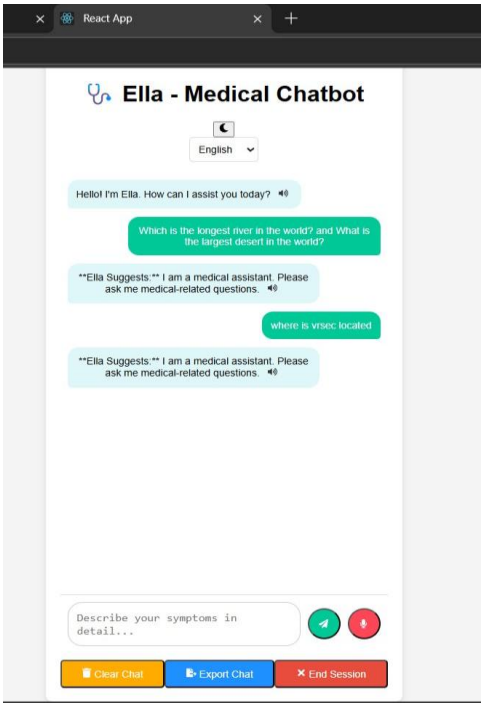


***Figure 4.7 Ella Rejecting Non-medical Queries***

The Figure 4.7 represents a chatbot that responds only to health or medical queries, redirecting non-medical questions to maintain focused and dedicated healthcare support.

## 4.5 OBSERVATIONS FROM THE WORK

### Model Performance and Response Quality

Based on the results, MedAlpaca-7B achieved the highest accuracy with the top BLEU (0.92) and ROUGE (0.94) scores. Mistral-7B also performed strongly, while GPT-3 showed the lowest accuracy among the models. In terms of speed, Meditron3 was the fastest, whereas the others had moderate latency. Overall, MedAlpaca-7B is the most accurate model, and Meditron3 provides the quickest responses

### User Interaction and Query Handling

The chatbot efficiently processed a wide range of medical queries, allowing users to describe symptoms in natural language and receive instant, AI-generated responses. The system also included fallback handling, which ensured that off-topic or non-medical questions were gently redirected. This kept the conversation focused and reinforced the chatbot's purpose as a healthcare assistant. Additionally, both the UI and the backend supported smooth interactions, offering a seamless experience for end-users.

## 4.6 SUMMARY

This chapter compares multiple models and shows that **MedAlpaca-7B delivers the highest accuracy**, while Meditron3 offers the fastest response time. Mistral-7B remains a strong overall performer, and GPT-3 ranks lowest in accuracy. The chatbot effectively handles user queries and maintains focused medical assistance throughout.

# CHAPTER-5

# CONCLUSION AND FUTURE WORK

This chapter presents the conclusions derived from the project, emphasizing the effectiveness, accuracy, and integration of the fine-tuned Mistral 7B model in the development of a medical diagnosis chatbot. It also outlines future directions for enhancing the chatbot's capabilities and usability across broader applications.

## 5.1 CONCLUSION

In this project, a medical diagnosis chatbot was developed using five LLM models—Mistral-7B, GPT-3, PMC-LLaMA, Meditron3-Gemma2-2B, and MedAlpaca-7B—fine-tuned and evaluated on a medical Q&A dataset. Among these, MedAlpaca-7B achieved the highest performance**,** with a BLEU score of 0.92 and ROUGE score of 0.94**,** making it the most accurate and fluent model for medical response generation. The other models contributed with varying strengths in speed, stability, and domain understanding.

The chatbot was integrated into an intuitive React.js interface supporting text, voice input, and medical query filtering, enabling smooth user interaction. Overall, the system combines the accuracy of advanced LLMs—led by MedAlpaca-7B—with a user-friendly design to deliver reliable and responsive medical assistance.

## 5.2 FUTURE WORK

1. **Utilizing Real-Time Medical Data:** Incorporating up-to-date real-time datasets into the model can significantly enhance its reliability and ensure that medical responses are aligned with the latest clinical guidelines and research findings.
2. **System Deployment and Accessibility:** To maximize usability, the chatbot should be deployed as a cloud-based solution, making it readily accessible to a broader audience, including healthcare professionals and general users across various devices and platforms

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